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From Accuracy to Defensibility: Estimation as Controlled Commitment in Time-Based IT Consulting. A Qualitative Case Study of Sopra Steria

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Abstract

This thesis challenges the dominant view that accuracy is the primary quality criterion for estimation in time-based IT consulting. Drawing on a qualitative single-case study of Sopra Steria, it argues that estimation in this setting is better evaluated by defensibility: whether an estimate can be justified as credible, manageable, and organisationally acceptable at the time of commitment.

Time-based consulting requires firms to commit consultants, competence, and price before scope, technical complexity, and customer decisions are settled. Weak estimates therefore do more than produce inaccurate forecasts: they distort staffing, erode margins, and damage customer relationships. Based on six semi-structured interviews with professionals involved in commercial estimation, staffing, and delivery-related decision-making, the thesis develops the concept of controlled commitment.

Controlled commitment describes how decision-makers commit enough to make an opportunity actionable while limiting exposure to unclear scope, unstable customer needs, scarce competence, and weak commercial conditions. The thesis shows that assignments are themselves partly shaped by estimation, that time-based contracts relocate rather than remove supplier commitment, and that staffing proposals function both as competence claims and risk allocations. By repositioning defensibility as the operative criterion for quality, the thesis challenges the assumption that estimated and actual effort are commensurable measures of estimation quality in knowledge-intensive, time-based work.

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Chapter 1

Introduction

In time-based IT consulting projects, decision-makers must estimate and justify resource and competence needs before the work is fully understood.

A customer may know that a problem must be solved, but not yet know the exact scope, technical complexity, dependencies, or competence required. The consulting firm must still decide which roles are needed, whether suitable consultants are available, what the work is likely to cost, and whether the organisation is willing to commit to an offer. In these early stages, estimation is therefore not only about predicting hours or costs. It is a decision-making process in which incomplete customer needs are translated into organisationally defensible staffing commitments.

This thesis studies how decision-makers estimate and justify resource and competence needs in time-based IT consulting projects. The study is a qualitative case study of Sopra Steria and examines estimation as an organisational practice shaped by judgement, uncertainty, resource availability, customer expectations, and commercial considerations. The thesis argues that resource and competence estimates are shaped by the organisational and commercial contexts in which they are produced and used.

The thesis addresses an underexplored part of project-based consulting practice by examining estimation as more than a technical forecasting task. It shifts attention to how resource and competence estimates are constructed, justified, and used as organisational commitments under uncertainty.

This chapter introduces the study's background, context, objectives, questions, contribution, and structure.

1.1 Background and Motivation

Estimation is central to software and IT projects because early assessments influence planning, staffing, pricing, delivery expectations, and commercial risk [11]. In time-based IT consulting projects, these assessments are especially important because the supplier is paid for consultants' time and competence rather than for a predefined technical product. The supplier must therefore estimate not only how much work may be required, but also what kind of competence the project needs.

Time-based IT consulting is knowledge-intensive work. Value creation depends largely on employees' knowledge, skills, experience, and judgement in practice [22]. Competence is therefore part of the offer itself.

A project estimate may involve decisions about roles, seniority, technical expertise, domain knowledge, customer-facing ability, team composition, and consultant availability. These assessments are not purely technical or organisational, but depend on decision-makers' experience, judgement, and interpretation of the project context. Estimation is therefore a situated judgement process shaped by how decision-makers assess and justify appropriate resource and competence needs.

These decisions matter for both consulting firms and customers. For consulting firms, weak estimation may lead to unrealistic staffing assumptions, delivery problems, margin pressure, or the loss of commercially attractive opportunities. For customers, poor estimates may reduce cost predictability, weaken delivery quality, and damage trust in the supplier relationship. Estimation therefore affects both internal planning and supplier-customer collaboration [7].

The topic is also relevant from a research perspective. Existing research provides important knowledge about software effort and cost estimation [11], project uncertainty [23], and resource allocation [14]. However, less is known about how these issues interact when decision-makers estimate and justify concrete resource and competence needs before or during a commercial commitment. This thesis therefore shifts attention from estimation as a technical forecasting task to estimation as the construction of organisational commitment under uncertainty.

This creates a need to study estimation as a situated decision-making process. In time-based IT consulting projects, decision-makers must connect customer needs, delivery assumptions, competence requirements, resource avail-

ability, cost, and commercial considerations. A better understanding of this process can provide insight into how consulting organisations make early project commitments when both the customer need and the available solution remain partly uncertain.

1.1.1 Sopra Steria

Sopra Steria is a European consulting company specialising in digital transformation, technology consulting, and innovation services. The company delivers advisory and implementation services in areas such as software development, data analytics, and IT infrastructure for both public- and private-sector clients [18].

In Norway, Sopra Steria is one of the largest IT consulting companies, with approximately 3,300 employees across offices in Oslo, Stavanger, Bergen, Trondheim, and Kristiansand. The Norwegian organisation reported revenue exceeding NOK 5.9 billion in 2024. Globally, Sopra Steria Group employs around 51,000 people across more than 30 countries and is headquartered in Paris [18].

Sopra Steria is a relevant case because much of its work is organised through project-based consulting engagements. These engagements require decision-makers to connect customer needs with roles, competence profiles, seniority levels, consultant availability, cost, and commercial commitment. The case therefore provides an appropriate setting for studying how resource and competence needs are estimated and justified in time-based IT consulting projects.

1.2 Research Objectives and Questions

This study aims to understand how decision-makers estimate and justify resource and competence needs in time-based IT consulting projects, and how they construct organisational commitments under uncertainty.

Rather than treating estimation primarily as a technical task of forecasting effort or cost, the study examines estimation as an organisational decision-making process in which decision-makers use judgement and experience to interpret incomplete information, make assumptions, and justify commitments under uncertainty.

The problem addressed in this thesis is the limited understanding of how estimation functions in practice as a basis for organisational commitment un-

der uncertainty. While prior research has examined estimation accuracy, uncertainty, and judgement, less is known about how these elements are combined when decision-makers construct resource and competence commitments in real organisational settings.

This problem is important because estimates in time-based consulting projects do more than describe expected work. They shape staffing decisions, delivery planning, pricing, risk assessment, and customer expectations, and influence the organisation's willingness to commit to an offer. Understanding these processes is therefore critical to explaining how organisations can act under uncertainty without committing beyond what can be justified.

Based on this problem, the study addresses the following research questions:

- How are resource and competence needs estimated and justified in time-based IT consulting projects?
- How is uncertainty handled and incorporated in these estimation processes?
- How are trade-offs between competence requirements, resource availability, cost, and commercial considerations balanced and justified?

Together, these questions guide the study by focusing on estimation as an organisational practice through which commitments are constructed, uncertainty is managed, and organisational and commercial constraints are balanced.

1.3 Approach and Contributions

This thesis uses a qualitative case study of Sopra Steria to examine estimation in time-based IT consulting projects. The empirical material consists of six semi-structured interviews with participants involved in sales, staffing, project planning, commercial estimation, customer responsibility, and delivery-related decision-making.

The study contributes to existing research in three ways.

- First, it examines estimation in an underexplored empirical context: time-based IT consulting projects.
- Second, it conceptualises resource and competence estimation as the construction of organisational commitment under uncertainty.
- Third, it provides practical insight for managers involved in the estimation and project commitment process.

1.4 Outline

The remainder of this thesis is structured as follows. Chapter 2 reviews relevant literature and positions the study within research on software project estimation, uncertainty, commercial influence, and resource allocation. Chapter 3 develops the theoretical and analytical framework for interpreting the empirical material. Chapter 4 describes the research design, data collection, data analysis, trustworthiness, and limitations of the study. Chapter 5 presents the empirical findings from the interviews, organised around three main themes. Chapter 6 discusses the findings in relation to the research questions, the analytical framework, and outlines implications for Sopra Steria. Finally, Chapter 7 summarises the thesis, highlights its contributions, and suggests directions for future research.

Chapter 2

Literature review

This chapter reviews research relevant to how resource and competence needs are estimated and justified in time-based IT consulting projects. The purpose is to examine what existing research explains about estimation, uncertainty, and staffing decisions, and to identify limitations in how these are understood in relation to customer-facing resource commitments. The chapter brings together three research streams.

Software estimation research explains how estimates are produced and how they are influenced by judgement and commercial conditions. Research on uncertainty explains how estimates are formed under incomplete information and changing assumptions. Research on resource allocation and project-based organisations explains how staffing decisions are shaped by availability, competing priorities, and organisational constraints.

Taken together, these perspectives suggest that estimation in time-based IT consulting projects extends beyond technical forecasting. It also involves constructing a defensible organisational commitment under uncertainty. The following sections present the literature search strategy, discuss each research stream, and synthesise them into a research gap.

2.1 Literature search and selection

The literature search was structured to identify key studies relevant to the themes of estimation, uncertainty, competence, staffing, and decision-making in project-based and software-related contexts. The search was not conducted as a system-

atic literature review, but as a focused review intended to synthesise relevant research and inform the study’s theoretical framework [17].

Table 2.1 summarises the literature search strategy, including databases, search terms, selection criteria, and the screening process.

Table 2.1: Overview of literature search strategy

Element	Description
Databases	Scopus, Scopus AI, Google Scholar, Research Rabbit, Oria
Search period	February 2026 to April 2026
Example search terms	“software project estimation”, “IT consulting”, “resource estimation”, “competence needs”, “staffing decisions”, “uncertainty in estimation”
Inclusion criteria	Peer-reviewed studies addressing estimation, uncertainty, competence, staffing, or decision-making in project-based or software-related contexts
Exclusion criteria	Studies focused primarily on algorithmic or technical estimation models without relevance to organisational decision-making or staffing practice
Selection process	Titles and abstracts were screened first. Relevant studies were then read in full, followed by backward and forward citation tracking.

2.2 Estimation as judgement in commercial settings

Previous research on software project estimation has mainly examined how effort and cost estimates are produced, improved, and evaluated. In their systematic review of 304 software cost estimation studies, Jørgensen and Shepperd [11] demonstrate that the field has developed around estimation methods, datasets, research designs, and accuracy evaluation. They classify the reviewed studies by approach, including regression-based models, analogy-based estimation, function points, work breakdown, classification and regression trees, simulation, and expert judgement. Taken together, this literature provides important insights into how estimates can be generated and assessed, particularly regarding effort,

cost, and accuracy.

At the same time, the estimation literature emphasises that estimation is not only a technical or model-based activity. In his review of expert estimation studies, Jørgensen [8] shows that expert judgement remains central to software development estimation, and that formal models have not consistently been shown to be superior across contexts. The review further indicates that judgment-based estimation can be improved through practices such as using historical data, producing independent estimates, justifying and critiquing estimates, using checklists, assessing uncertainty, and providing feedback on estimation accuracy. Even formal estimation models require judgement when estimators assess project-specific input factors such as complexity, team skill, and customer requirements [10]. Taken together, these studies suggest that estimation involves both interpretation and calculation.

This insight is particularly relevant for the present study, as resource and competence estimation requires decision-makers to interpret incomplete customer needs and translate them into assumptions about roles, skills, seniority, and capacity. As prior research indicates, such decisions cannot be derived mechanically from project information, but depend on experience-based judgement [10]. They rely on assessments of what the project appears to require, which competence profiles seem credible, and which assumptions can be communicated to the customer.

In addition, research shows that software estimates are shaped by commercial and customer-facing contexts. Customer expectations about total project cost can influence judgment-based estimates of software development effort, even when estimators are instructed not to rely on such expectations [12]. Similarly, another study suggests that the format of software project bidding processes can influence submitted bids and client selection [9]. These findings indicate that estimates and bids are not neutral representations of expected work, but may reflect customer expectations, bidding formats, price-to-win considerations, and broader commercial framing.

This commercial dimension is particularly relevant in time-based IT consulting contexts. The supplier is not only estimating how much work may be required, but may also need to propose what type of competence the customer should buy, how the delivery should be staffed, and under what assumptions the supplier is willing to commit. In line with prior research, estimation can therefore be under-

stood as part of a broader commercial commitment process [12, 9]. The estimate must be credible to the customer, feasible for the supplier, and defensible as a basis for delivery.

Overall, the software estimation literature highlights two key insights. First, estimation depends on judgement, even when supported by formal methods or historical data. Second, estimation is influenced by commercial and customer-facing conditions. However, most studies focus primarily on effort, cost, price, and accuracy [11], offering limited insight into how consulting firms estimate and justify resource and competence requirements for customer-facing offers.

This limitation motivates the analytical concept of estimation basis. In this thesis, the estimation basis refers to the information, assumptions, experience, historical references, and forms of judgement that decision-makers rely on when constructing an estimate. The concept highlights the foundation on which a consulting firm makes a customer-facing commitment when information is incomplete.

2.3 Uncertainty, assumptions, and conditional commitments

Uncertainty is recognised in the project management literature as a central characteristic of project estimation, as estimates are typically produced before all relevant information is available. Ward and Chapman [23] argue that uncertainty should not be understood only as identifiable risk events or threats, but more broadly as uncertainty related to the basis of estimates, assumptions, objectives, priorities, design, logistics, and relationships among project actors. This broader understanding shifts attention from whether estimates are accurate to how estimates are constructed when important conditions remain unsettled.

Research further indicates that uncertainty may affect both the customer and supplier sides of a project. In time-based IT consulting projects, customer needs, scope, technical complexity, and responsibilities may be unclear during the offer preparation phase. At the same time, the supplier may face uncertainty regarding future resource availability, competence profiles, and the degree of commitment it is willing to make, reflecting uncertainty in resource allocation under dynamic project conditions [14]. As a result, estimation involves not only deter-

mining what the project may require, but also deciding how much uncertainty the organisation is willing to absorb in its commitments.

Several studies highlight that uncertainty influences both the content and the justification of estimates [23, 8]. It shapes which assumptions are made, how explicitly they are communicated, and what reservations or qualifications are included. Estimates may therefore function as conditional commitments, defining what is currently understood about the project while acknowledging what remains uncertain or subject to change.

In this thesis, uncertainty is treated not only as a factor that affects estimation accuracy, but as something decision-makers actively manage. This perspective is consistent with prior research suggesting that uncertainty requires broader management attention and flexibility in dynamic project environments [23, 14]. This is particularly important in time-based consulting, where customer expectations may shape estimates despite evolving needs and incomplete information [12].

While the uncertainty literature provides important insight into how estimates are based on assumptions and incomplete knowledge [23, 14], it offers less detail on how such uncertainty is incorporated into concrete resource and competence commitments. In particular, it offers only a limited understanding of how decision-makers determine which assumptions, reservations, and competence claims to include in customer-facing offers.

This limitation motivates the analytical concept of uncertainty handling. In this thesis, uncertainty handling refers to the use of assumptions, reservations, qualifications, and flexibility to enable organisational commitments despite incomplete information. The concept focuses on how uncertainty is bounded and communicated during the estimation process.

2.4 Competence and staffing as organisational trade-offs

Research on project-based organisations shows that staffing decisions are embedded in organisational constraints. Studies of project-oriented companies highlight that projects are temporary and dynamic, employees may hold multiple organisational roles, and resource configurations change over time [6]. As a result, staffing is not only an administrative task, but a central part of strategic human

resource management.

This perspective is particularly relevant in time-based IT consulting, where estimation is closely linked to staffing decisions. However, this thesis distinguishes between estimation and allocation. Allocation refers to how available personnel are assigned to projects [14], whereas estimation concerns how resource needs are defined and justified before or during commercial commitment [11]. Although these processes are interrelated, they involve different types of decision-making.

Within this context, competence is understood as more than a generic resource attribute. It refers to roles, skills, seniority levels, domain knowledge, and the credibility required to deliver the project. Estimating competence involves judging what expertise is needed, how specialised it should be, and how these decisions influence cost, availability, risk, and customer confidence.

Studies of resource allocation and project-based organisations emphasise that staffing decisions are shaped by organisational constraints, availability, prioritisation, and changing project conditions [6, 14]. Highly specialised or senior competence may reduce delivery risk and strengthen customer confidence, but may increase cost or limit availability. More general profiles may be easier to staff and more commercially attractive, but may require stronger assumptions or more limited commitments. Estimation therefore involves balancing project requirements, organisational feasibility, and commercial considerations.

While the resource allocation literature provides important insights into how staffing decisions are constrained by organisational factors, it focuses primarily on resource allocation and reallocation [14]. It offers less insight into how resource and competence requirements are initially constructed and justified in customer-facing offers.

This limitation motivates two analytical concepts. The first is competence justification, which refers to how requirements for roles, skills, seniority, and domain knowledge are explained and defended in relation to customer needs. The second is commercial and strategic trade-offs, which refers to how decision-makers balance competence requirements, staffing feasibility, cost, availability, risk, and customer expectations when constructing a defensible offer.

2.5 Synthesis and research gap

The reviewed literature shows that estimation in project-based work is shaped by judgement, uncertainty, and organisational constraints. Software estimation research highlights the role of expert judgement, historical information, and commercial influences in shaping estimates [8, 12, 9]. Research on uncertainty shows that estimates are based on assumptions and incomplete knowledge [23], while studies of resource allocation demonstrate that staffing decisions are constrained by availability, competing demands, and temporary project structures [6, 14].

Taken together, these perspectives suggest that estimation in time-based IT consulting projects extends beyond forecasting expected work. Estimation also contributes to the development of a defensible customer-facing offer under uncertainty. Such an offer defines not only expected effort, but also the competence offered to the customer, the assumptions underlying the proposal, and the level of organisational commitment the supplier is prepared to make.

However, existing research provides limited insight into how these elements are integrated in practice. While prior studies have examined estimation methods, uncertainty, and resource allocation separately, less is known about how decision-makers combine these considerations when constructing concrete resource and competence commitments before a project is fully specified.

This thesis addresses this research gap by examining how decision-makers estimate and justify resource and competence needs in situations characterised by incomplete information about customer needs. The study focuses on how such assessments are shaped by judgement, experience, uncertainty, resource availability, and commercial considerations. In doing so, estimation is conceptualised as a situated organisational decision-making process through which consulting firms construct and justify commitments under uncertainty.

Chapter 3

Theoretical framework

This chapter presents the theoretical framework of the study. Drawing on literature on project estimation, uncertainty, competence, and knowledge-intensive services, estimation is understood as an organisational practice in which uncertain customer needs are translated into staffing assumptions, competence requirements, and commercial commitments.

On this basis, four analytical concepts are developed: estimation basis, uncertainty handling, competence justification, and commercial and strategic trade-offs. These concepts are selected because they enable an analysis of how decision-makers construct and justify resource and competence needs under conditions of uncertainty in practice. Together, they provide a framework for examining how assessments are shaped, how uncertainty is managed, and how trade-offs are balanced when organisational commitments are established [7, 1].

The selection of these concepts is guided by the study's aim to understand estimation as an organisational decision-making process. Together, they capture how decision-makers construct, justify, and balance resource and competence needs under conditions of uncertainty, thereby providing an analytically relevant lens for examining estimation in practice.

3.1 Estimation as construction of organisational commitment

Time-based IT consulting projects can be understood as knowledge-intensive work in which value creation depends on employees' knowledge, skills, and judgement. At the same time, the quality of such work may be difficult for clients and non-experts to assess in advance [22]. This creates a fundamental challenge for estimation: decision-makers must justify staffing needs and competence requirements before the value of the work can be demonstrated.

This implies that estimation is not primarily a technical calculation, but a process through which decision-makers construct and justify organisational commitments under conditions of uncertainty [22, 1].

In this context, estimation is not only a calculation of expected work but also a process by which an uncertain assignment becomes organisationally actionable. Decision-makers interpret the customer's problem, define the likely scope of work, identify relevant roles and competence profiles, and assess resource availability under uncertainty [1], while also considering whether the proposed setup can be commercially justified [7].

Research on project bidding supports this understanding. Jørgensen and Carelius [7] show that bidding involves not only estimating effort, but also considering uncertainty, margins, client budgets, competitive pricing, and long-term strategic factors. Estimates are therefore not neutral descriptions of work, but commitments shaped by what the organisation is willing and able to offer.

From this perspective, estimates can be understood as provisional commitments that make a project actionable. The analytical focus shifts from whether an estimate is accurate to how it becomes credible, defensible, and commercially acceptable under conditions of uncertainty [7, 1].

3.2 Estimation basis

The first analytical concept is estimation basis, which refers to the information, assumptions, experience, and judgement used to construct an understanding of the assignment. It focuses on what decision-makers treat as known, what they infer, and what remains uncertain [1]. In early project phases, information about

customer needs, technical conditions, and delivery constraints is often incomplete [7]. The estimation basis therefore combines explicit inputs, such as requirements, deadlines, and role expectations, with interpretive assumptions about workload, complexity, and resource needs [1, 7].

The basis for estimation shapes the definition of the project [7]. It influences which competence profiles appear necessary, which uncertainties become visible, and which trade-offs become relevant [1, 7]. For example, framing a project as technically complex may justify using senior specialists, whereas a commercially constrained project may lead to leaner staffing assumptions.

The concept therefore directs attention to how decision-makers move from incomplete input to concrete staffing expectations. It provides the starting point for analysing uncertainty handling, competence justification, and trade-offs.

3.3 Uncertainty handling

The second concept is uncertainty handling, which refers to how decision-makers interpret and manage incomplete or ambiguous information during estimation.

Project research shows that uncertainty is not limited to identifiable risks but includes unclear requirements, assumptions, stakeholder expectations, and resource availability. In time-based IT consulting, uncertainty may affect both customer needs and supplier capabilities [1].

Decision-makers do not necessarily eliminate uncertainty. Instead, they transform it into manageable forms, such as assumptions, reservations, buffers, or flexible staffing arrangements. Uncertainty may be reduced through additional information or experience, but it may also be accepted when strategic or commercial considerations support commitment [1, 7].

The concept therefore focuses on how uncertainty is made manageable enough to support action. It examines how decision-makers interpret uncertainty, which aspects they prioritise, and whether to reduce, compensate for, or accept it.

3.4 Competence justification

The third concept is competence justification, which concerns how resource and competence needs are explained and made legitimate.

In time-based consulting, estimation involves not only how much work is required but also the kinds of people needed. Competence includes technical skills, domain knowledge, and seniority [13], as well as the ability to handle uncertainty and interact with clients [1].

Competence can be understood both as a delivery resource and as an organisational capability [13]. It may be justified in terms of delivery quality, risk reduction, client expectations, or long-term organisational development [13, 1]. These justifications may align, but often involve tension.

Because competence is partly intangible, it must be made visible and defensible [22]. Decision-makers translate vague customer needs into roles, profiles, and levels of seniority that are both internally understood and accepted by the client [1].

The concept therefore focuses on how competence requirements are justified in relation to task complexity, uncertainty, client expectations, and organisational interests.

3.5 Commercial and strategic trade-offs

The fourth concept is commercial and strategic trade-offs, which refers to the process of balancing professional, organisational, and commercial considerations.

Higher competence may strengthen delivery credibility but increase cost or reduce flexibility [13]. More available resources may improve feasibility but increase delivery risk [1]. Strategic considerations such as entering a new market or developing competence may further influence decisions [7, 13].

Estimates can therefore be understood as negotiated compromises rather than purely technical outcomes. They reflect what the organisation considers professionally defensible, commercially viable, and practically possible. This makes commercial and strategic trade-offs central to understanding how resource and competence commitments are constructed in time-based IT consulting projects.

Chapter 4

Methodology

This chapter explains how the study was designed and conducted. The study adopts a qualitative, interpretive single-case design to understand how decision-makers estimate and justify resource and competence needs in time-based IT consulting projects. Empirically, it draws on six semi-structured interviews with decision-makers at Sopra Steria involved in commercial, staffing, planning, and delivery-related work.

4.1 Research design

This study adopts a qualitative single-case study design. A qualitative approach is appropriate because the research questions concern how decision-makers understand, justify, and handle estimation in practice. These processes involve judgment, interpretation, uncertainty, and organisational context, which makes qualitative methods suitable for studying participants' reasoning and experiences [3].

Sopra Steria provides the organisational context for the case study. A case study design is suitable for studying a contemporary phenomenon in its real-life context, especially when the boundaries between the phenomenon and its context are unclear [24].

The case is defined as the practice of estimating and justifying resource and competence needs in time-based IT consulting projects within Sopra Steria. The unit of analysis is the decision-making process through which resource and competence needs are estimated and justified [24].

Sopra Steria was selected because resource and competence decisions are cen-

tral to project delivery in knowledge-intensive IT consulting. It also provided access to participants with direct experience of the relevant estimation decisions.

The aim of the case study is analytical insight rather than statistical generalisation, meaning that the study seeks to generate contextually grounded understanding that may be analytically transferable to similar settings [24].

Survey and observational approaches were considered but rejected because they would have provided less insight into how estimation is reasoned and justified in practice. Semi-structured interviews were therefore selected as the primary data collection method [3].

4.1.1 Case access and researcher position

The researcher's employment at Sopra Steria provided access to participants with direct experience of estimation-related decision-making, which would have been difficult to study from the outside.

This insider position supported contextual understanding and may have helped establish trust during interviews. The ethical implications of the researcher's insider position are discussed in Section 4.4.

4.2 Interviews

Semi-structured interviews were used as the primary data collection method. This method was suitable because estimation in IT consulting is difficult to observe directly and often depends on experience-based reasoning, informal judgement, and contextual knowledge [3].

4.2.1 Selection of participants

Participants were selected through purposive sampling. The aim was to recruit individuals with experience relevant to the study's research questions [3]. The selection focused on professionals involved in estimation-related decision-making in consulting engagements, including sales, staffing, project planning, competence assessment, delivery, and customer-facing advisory work.

Six professionals with decision-making responsibilities participated in the study. The participants were based at Sopra Steria's Oslo and Stavanger offices and rep-

resented three organisational areas: Applications, Digital Platform Services, and Advisory. The sample included participants with experience from commercial estimation, proposal work, staffing and resource coordination, project delivery, competence assessment, personnel responsibility, and customer-facing advisory work. This provided access to diverse perspectives on estimation-related decision-making across different organisational roles.

The labels P1–P6 are used consistently throughout the thesis when presenting findings, and the mapping between these labels and the interviewees is kept confidential. Potential participants were contacted through Microsoft Teams. In total, eight potential participants were contacted. Six participated, while two did not participate.

4.2.2 Interview guide

The interview guide was developed from the research questions and relevant literature on project estimation and decision-making in knowledge-intensive organisations. It was organised thematically to move the conversation from descriptive accounts of practice to more reflective discussions of decision-making.

The interviews began with participants' professional backgrounds and involvement in estimation. Participants were then asked how estimation processes typically unfold in time-based consulting projects.

The central part of the guide followed the three research questions. It addressed how resource and competence needs are estimated and justified, how uncertainty is experienced and handled, and how trade-offs are balanced when competence requirements, resource availability, cost, and commercial considerations conflict.

Open-ended questions encouraged participants to provide examples from practice. Follow-up questions were used to clarify answers and explore emerging topics in more depth. This structure supported both focus on the research objectives and flexibility for participants to describe their experiences in their own terms [3]. The complete interview guide is presented in Appendix C.

4.2.3 Interview procedure

The interviews were conducted between March and April 2026. They lasted 49 minutes on average. Interviews with participants in Stavanger were conducted

in person, while those in other locations were conducted digitally via Microsoft Teams.

All interviews were conducted in Norwegian. This was the most natural working language for both the researcher and the participants, allowing them to use familiar organisational terminology.

The same semi-structured interview guide was used in all interviews. The main questions were asked to all participants, but the order, wording, and follow-up questions varied with the flow of the conversation [3].

The first interview also served as a pilot interview. As no substantial changes were required, the guide was retained, and the pilot interview was included in the dataset.

Relevant excerpts were translated from Norwegian to English for presentation in the thesis. The translation prioritised meaning while making the quotations readable in English.

4.3 Data analysis

The interview data were analysed using thematic analysis. The analysis focused on identifying patterns in the empirical material and interpreting these patterns in light of the theoretical framework.

This method is suitable for identifying, analysing, and reporting patterns in qualitative data [2] and aligns with the aim of this study, which examines how decision-makers describe, experience, and justify estimation in time-based IT consulting projects.

The analysis followed the six-phase framework proposed by Braun and Clarke [2], used as a flexible analytical guide. The analysis moved iteratively between the empirical material and the theoretical framework. Coding was guided by the research questions and concepts from the theoretical framework, while also allowing additional patterns to emerge from the data.

The analysis began with repeated reading of the manually reviewed transcripts. Relevant text segments were coded in NVivo, focusing on estimation practices, uncertainty, competence assessments, trade-offs, and decision-making processes. Inductive codes included customer ambiguity, use of prior experience, assumptions, phased delivery, and BCR approval.

The analysis was documented through transcripts, coded material, code groupings, and theme-development notes, creating an audit trail from the empirical material to the final themes [15].

To illustrate how the analytical process moved from empirical material to final themes, Appendix D provides an example of the progression from an interview excerpt to an initial code, a descriptive category, and a final theme. This supports transparency in the analytical process without placing extensive coding details in the findings chapter.

This approach enabled the study to examine how estimation practices are constructed, justified, and shaped within organisational and commercial contexts.

4.4 Ethical considerations

The study followed institutional research ethics guidelines at the University of Stavanger [21] and national research ethics guidelines for the social sciences and humanities [19]. As the study involved the processing of personal data, it was reported to the Norwegian Agency for Shared Services in Education and Research (Sikt) before data collection. The notification form, reference number 290155, is included in Appendix A.

Participation was voluntary and based on informed consent. Before the interviews, all participants received written information about the study's purpose, data handling, confidentiality, and their rights. They were informed that participation was voluntary and that they could withdraw from the study in accordance with the conditions described in the consent form. The information sheet and consent form provided to participants are included in Appendix B.

Data processing and storage complied with the EU General Data Protection Regulation 2016/679 (GDPR) [4] and University of Stavanger guidelines [21]. Recordings and transcripts were stored in approved systems, with access restricted to the researcher. The recordings were transcribed using Nettskjema's Autotekst service [20]. The transcripts were then manually reviewed and anonymised before analysis by removing or generalising names of participants, colleagues, clients, and specific projects.

To reduce the risk of identification, participant characteristics are reported only in aggregate. The study avoids combining department, location, role, or de-

cision context at the participant level. Participants are referred to as P1–P6, and the mapping between participant labels and interviewees is kept confidential.

The researcher’s employment within the case organisation raised ethical considerations regarding role duality, organisational proximity, and potential power asymmetries. Such proximity may influence participants’ responses [19]. To reduce this risk, participants were selected from outside the researcher’s immediate professional context, the interviews used open-ended questions, and participants were assured that their responses would remain confidential.

4.5 Trustworthiness in qualitative research

The study uses credibility, transferability, dependability, and confirmability to assess trustworthiness in qualitative research [5, 16].

Trustworthiness was strengthened through purposive sampling, semi-structured interviews, audio recordings, reviewed transcripts, systematic coding, and interview excerpts.

4.5.1 Credibility

Credibility concerns whether the findings plausibly represent participants’ perspectives and experiences [16]. The study strengthened credibility by selecting participants with direct experience of estimation-related decisions and by using semi-structured interviews.

The use of recordings, transcription, and manual transcript review reduced the risk of misrepresenting participants’ statements. During analysis, interpretations were grounded in coded interview material, and Chapter 5 uses excerpts to show how the final themes are supported by the data.

4.5.2 Transferability

Transferability concerns whether the findings may be relevant to other contexts [16]. As this study is based on a single organisational case, it does not aim to generalise statistically. Instead, transferability is supported by providing contextual information about the case organisation, participants, and data collection process.

4.5.3 Dependability

Dependability concerns the transparency and consistency of the research process [16]. The study addressed dependability by documenting the main methodological decisions, including the case design, participant selection, interview guide, interview procedure, and thematic analysis.

4.5.4 Confirmability

Confirmability concerns whether the findings are grounded in the empirical material rather than in the researcher's personal assumptions [16]. The study addressed confirmability by linking the findings to coded interview material and by using interview excerpts to illustrate the final themes.

As discussed in Section 4.1.1, the researcher's insider position created both access advantages and a risk of bias.

4.6 Limitations

The study has several methodological limitations that affect how the findings should be interpreted.

First, the study is based on a single organisational case. The findings may therefore reflect Sopra Steria's specific structure, culture, client base, and staffing practices. They cannot be statistically generalised across IT consulting organisations. The aim is instead analytical generalisation to similar knowledge-intensive and project-based consulting contexts [24].

Second, the study relies primarily on six semi-structured interviews. The findings therefore reflect how participants describe and interpret estimation practices, rather than direct observation of estimation meetings or documentary evidence of specific project decisions. Other actors involved in estimation processes may have provided different perspectives.

Third, the researcher's employment at the case organisation may have influenced the research process. As discussed in Section 4.1.1, the insider position supported access and contextual understanding, but also introduced a risk of bias and taken-for-granted assumptions. The findings should therefore be read as interpretations developed from participants' accounts within a familiar organisa-

tional context, not as a detached external audit of the organisation's estimation practices.

Fourth, the interviews were conducted in Norwegian and presented in English. Translation may have affected nuance, especially where participants used informal expressions, organisational terminology, or context-specific descriptions. The translation of the excerpts aimed to preserve meaning while making the quotations readable in English.

Fifth, automated transcription introduced a potential source of error. Manual review and correction of all transcripts reduced this limitation before coding and analysis.

Finally, confidentiality limited access to internal project documentation and commercial material. The study therefore relies mainly on interviews rather than triangulation with internal documents. The findings should consequently be interpreted as an analysis of participants' accounts, not as a direct audit of specific estimation decisions or project documents.

Chapter 5

Findings

This chapter presents the findings from the thematic analysis. The central finding is that estimation in time-based IT consulting projects is not mainly a technical calculation of hours, roles, or costs. It is an organisational process for turning uncertain customer needs into proposals that can be staffed, priced, reviewed, and communicated.

This pattern can be understood as controlled commitment. Decision-makers commit enough to make an opportunity actionable, but not so much that the organisation becomes overexposed to unclear scope, unstable customer needs, unavailable competence, or weak commercial conditions. The estimate does not remove uncertainty. It makes uncertainty manageable by translating it into roles, assumptions, ranges, reservations, phases, and staffing choices.

The three themes were developed through the thematic analysis described in Chapter 4. Interview excerpts were coded for recurring patterns in estimation practices, uncertainty, competence assessment, trade-offs, and organisational decision-making. These patterns were then grouped into broader categories and refined into the final themes presented in this chapter. Selected excerpts are used throughout to illustrate how the themes are grounded in the empirical material.

The findings are organised around three themes. First, participants translated incomplete customer needs into organisationally defensible staffing commitments. Second, they bounded uncertainty by qualifying, limiting, and sequencing the supplier's commitments. Third, they negotiated staffing proposals based on competence fit, availability, cost, margin, and strategic value.

Table 5.1 summarises the themes and sub-themes.

Table 5.1: Findings structure: themes and sub-themes

Main theme	Sub-themes
Translating customer needs into staffing commitments	• Using prior experience without a fully systematic data basis • Mapping customer needs to roles • Making estimates organisationally actionable
Managing uncertainty through assumptions and phased commitment	• Uncertainty from unclear scope and customer context • Setting bounds through assumptions and ranges • Phased delivery to reduce uncertainty
Negotiating staffing: fit, availability, and strategic trade-offs	• Balancing competence, availability, and customer fit • Using seniority and team composition • Balancing cost, margin, and strategy

5.1 Translating customer needs into staffing commitments

Estimation often began before the customer's need was fully specified. Decision-makers could therefore not allocate resources based on predefined requirements. They first had to interpret the customer problem and translate it into organisational categories such as roles, competence profiles, seniority levels, hours, rates, and delivery responsibilities. This translation can be understood as part of the basis for estimation, where decision-makers construct a workable understanding of the assignment from incomplete and ambiguous input. The process involved not only identifying what the customer needed, but making that need actionable within the supplier organisation.

5.1.1 Using prior experience without a fully systematic data basis

Prior experience provided decision-makers with an important starting point, but participants described this knowledge base as only partially systematic. Previous projects, earlier offers, templates, reference numbers, and experienced colleagues helped form an initial view of the work. However, this knowledge was often drawn from memory, informal comparison, and expert judgement rather than from structured historical data.

P6 explained:

“We use earlier projects a lot ... We often use top-of-mind experience ... It could probably have been much more data-driven, since there is no real link between previous offers and what the actually invoiced hours were afterwards.”

For recurring work, reference numbers made estimation easier. P3 explained:

“We use a lot of reference numbers ... X number of hours to migrate a server, or X number of hours to migrate an application.”

However, historical experience could not simply be copied from one project to another. P4 noted:

“Even though we have done a lot many times before ... there is always a degree of individual adaptation, because the organisation, technical system, and activity landscape are not the same.”

Prior experience therefore worked as a reference point rather than a rule. This illustrates how the basis for estimation is constituted not only by formal data but also by experience-based judgement and organisational memory. It provided direction and legitimacy to early estimates, while still requiring interpretation and adaptation to the current context.

5.1.2 Mapping customer needs to roles

Customer requests rarely entered the estimation process as ready-made staffing requirements. Before estimating hours, roles, or seniority, decision-makers had

to interpret the customer's problem and identify the competence needed to solve it.

P2 explained:

"It can be extremely thin or very detailed. Then it is about interpreting what needs to be solved, and what type of competence is needed to solve it."

P1 described a similar situation:

"When customers ask us to make an offer, they are often unsure about what they actually need."

To manage this ambiguity, decision-makers mobilised internal expertise. P4 explained:

"I use many people outside the team... Based on your experience, what do you think is a good estimate?"

This process also reflects competence justification, as decision-makers translated uncertain customer needs into specific roles, profiles, and seniority levels that could be understood internally and justified externally. Estimation therefore had an advisory function: it not only responded to customer needs, but also helped define them.

5.1.3 Making estimates organisationally actionable

Internal review transformed situated judgements into organisational commitments. Through discussion and approval processes, estimates became collectively owned and legitimised.

P2 described BCR as follows:

"It is a formal meeting point with documentation, where you go through what you have planned to price and offer."

P2 also emphasised that the process created shared responsibility:

"We make a joint commitment ... We took joint responsibility."

Much of the groundwork occurred before formal review. P3 explained that estimates were discussed with project managers, architects, and specialists before they entered formal decision arenas. The review process therefore did more than validate a final number. It coordinated knowledge across roles and functions.

Internal review changed the status of the estimate. Before review, the estimate was a judgement made by a limited group. After review, it became something the organisation could stand behind.

This shows how estimation is involved not only in constructing an estimation basis but also in ensuring that the proposed commitment becomes organisationally acceptable. The process did not remove uncertainty, but made it collectively understood and accepted.

5.2 Managing uncertainty through assumptions and phased commitment

The second theme shows how decision-makers controlled the exposure created by staffing proposals. In time-based consulting, estimates had to support planning and customer decisions without creating unrealistic expectations of precision. This can be understood as uncertainty handling, in which decision-makers actively interpret and manage incomplete information to enable commitment.

5.2.1 Uncertainty from unclear scope and customer context

Uncertainty stemmed mainly from unclear scope and customer-side conditions. Participants mentioned changing requirements, access issues, internal decision-making, stakeholder availability, and dependencies outside the supplier's control.

P5 pointed to changing scope:

“Wavering customers create the most uncertainty.”

P6 similarly emphasised customer-side dependencies:

“It is all the internal factors at the customer that we do not control ourselves.”

The uncertainty participants described was therefore not only technical. It was also relational and organisational. It depended on the customer's ability to clarify needs, provide access, make decisions, involve stakeholders, and maintain direction. Such uncertainty could not be solved through technical estimation alone. It had to be managed through expectations, assumptions, and boundaries around the supplier's responsibility.

5.2.2 Setting bounds through assumptions and ranges

Participants managed uncertainty by making the conditions of the estimate explicit, which can be understood as a form of uncertainty handling. Assumptions, ranges, reservations, and scenario-based estimates clarified what was known, what remained uncertain, and what had to hold for the estimate to remain valid.

This was particularly important in time-based projects. Several participants distinguished time-based work from fixed-price or target-price projects, in which the supplier bears greater responsibility for delivering within agreed-upon cost or scope constraints. In time-based projects, estimates were often treated as indications rather than binding predictions.

P2 explained:

“When it is time-based ..., our responsibility is not there.”

This created a tension between contractual and relational responsibility. Formally, time-based estimates were limited commitments. In practice, they still shaped customer expectations and could later require an explanation. P2 referred to this as implicit responsibility:

“You get what is called implicit responsibility.”

One way to manage this tension was to avoid false precision. Instead of presenting a single number, participants used ranges, scenarios, and conditions to show that the estimate depended on uncertain factors.

P3 described three-point estimation:

“We normally use a three-point estimate: best case, worst case, and mid case. We weigh the mid case four times ... It forces you to consider how well things could go, and how badly they could go.”

P4 described the same logic in practical terms:

“Three to five weeks, two to four months. If this happens, then we are probably at the lower end, and if this happens, then we are probably at the higher end.”

Reservations served a similar function. When important information was missing, participants used assumptions or reservations to clarify what had to be true for the estimate to remain valid. P6 explained:

“If we cannot get an answer to the question, then we simply have to make reservations.”

These practices bound uncertainty in different ways. Ranges expressed numerical uncertainty, scenarios explained how the estimate might change, and reservations clarified the conditions under which the estimate held. Together, they made uncertainty visible without requiring the supplier to resolve it fully before the project began.

5.2.3 Phased delivery to reduce uncertainty

Phased delivery bounded uncertainty by changing the timing of commitment. Discovery phases, proofs of concept, MVPs, FAST models, and agile breakdowns enabled decision-makers to replace a single large early commitment with a sequence of smaller, revisable commitments.

P4 described using a proof of concept before making a larger commitment:

“Let us make a proof of concept (POC), let us test it out. This is what is uncertain, and that is what we will find out ... we will come back with an updated estimate when we are finished with the first phase.”

The proof of concept was therefore not only a technical test. It also functioned as an estimation device. It limited the supplier’s initial commitment and generated information that could justify a later revision of the estimate.

P1 described a similar logic in relation to agile work:

“It does not make much sense to estimate something from A to Z.”

Phased delivery made estimation iterative rather than final. Instead of committing to the whole project while information was limited, the supplier and customer could make a smaller initial commitment, learn from the first phase, and reassess the larger opportunity.

5.3 Negotiating staffing: fit, availability, and strategic trade-offs

Staffing proposals were not derived directly from the estimated work, but emerged through commercial and strategic trade-offs. Although competence was usually the formal starting point, the final proposal emerged through negotiation between several organisational logics. Delivery logic prioritised the team best suited to solve the customer's problem. Utilisation logic pushed decision-makers to consider available capacity. Commercial logic required attention to rates, costs, and margins. Strategic logic could make lower immediate profitability acceptable if the customer relationship was valuable.

This negotiation was the third dimension of controlled commitment. The estimate had to describe a team that was technically credible, organisationally realistic, and commercially viable. The final staffing proposal was therefore not simply the ideal team, but the best defensible team under current constraints.

5.3.1 Balancing competence, availability, and customer fit

Competence defined the starting point for staffing, but availability shaped what could actually be offered. Decision-makers considered technical skills, domain experience, seniority, and customer fit when assessing who would suit an assignment. However, the best-matched consultant was not always available, and the available consultants were not always the strongest fit.

P2 explained:

“We start looking at what type of profiles we need, and then of course, who is available comes into the picture ... You always prefer to choose the ones who are available, because it is quick and easy.”

P1 was more direct:

“The first thing you think about in our industry is availability.”

Availability mattered because unused capacity meant lost billable time. At the same time, participants warned against letting availability dominate the staffing decision.

P6 cautioned against this:

“You always try not to staff the most available resource, but the one who is best suited to carry out the project.”

Competence also meant more than technical skill. Participants described customer fit, communication style, domain experience, personality, and working style when assessing suitability.

P5 distinguished formal competence from contextual fit:

“Even if that person has the right competence on paper, it is not certain that it is the right person for what the customer has asked for.”

The staffing estimate was therefore shaped by a tension between delivery fit and capacity utilisation. Delivery logic defined the right consultant as the person best suited to the customer’s problem. Utilisation logic defined the right consultant as someone whose available capacity could be converted into billable work. The final proposal reflected both logics.

5.3.2 Using seniority and team composition

Seniority and team composition helped decision-makers balance delivery quality, cost, and risk. Senior consultants were more expensive, but participants associated them with efficiency, experience, and lower delivery risk. Junior consultants could reduce costs and support internal competence development, but usually require support from more experienced colleagues.

P3 linked seniority and complexity:

“If it is a simple task, then we can typically use a graduate. But if you are building architecture in a data centre, then I may need a senior manager or a director.”

P4 described the cost-quality trade-off:

“The more experienced consultants cost much more in hourly rate, but they should also be able to deliver it in much less time and with better quality.”

Participants did not understand cost only as an hourly rate. A senior consultant could appear more expensive per hour, but still be justified if they were expected to deliver faster, make better decisions, or reduce the need for later correction. Conversely, junior consultants could make the proposal less costly, but only when the work was sufficiently structured or when senior support was available.

P6 described combining senior and junior consultants:

“You can either have a director with a high hourly cost, or you can add several juniors ... to reduce the cost.”

Seniority therefore functioned not only as a competence or price category, but as a way of distributing delivery risk. Senior consultants could handle ambiguity, make architectural decisions, and guide less-experienced consultants, while junior consultants could reduce costs when tasks were clearer. The seniority mix reflected assumptions about complexity, uncertainty, and the support the project would require.

5.3.3 Balancing cost, margin, and strategy

Commercial considerations shaped which staffing estimates were accepted, but profitability was not assessed solely at the project level. Participants distinguished between project-level and account-level commercial reasoning. At the project level, decision-makers considered cost, rates, margins, and expected delivery effort. At the account level, they assessed whether the customer relationship could generate future volume, strengthen strategic positioning, or provide access to additional work.

This illustrates how staffing decisions were shaped by commercial and strategic trade-offs, where short-term profitability was weighed against longer-term organisational value. As P5 explained:

“If we have to reduce the margin in order to win a case, but we believe that the customer can generate 300 consultants in projects for us, then we are willing to take the risk.”

In such cases, a lower immediate margin could be commercially rational if the opportunity had value beyond the specific project. Strategic considerations therefore influenced how commercial risk was interpreted. A low-margin estimate was not necessarily perceived as weak; rather, it could be understood as an investment in a broader customer relationship.

Participants also distinguished between delivery cost and sales price. As P3 stated:

“One thing is what the project costs; another thing is what we sell it for.”

P6 similarly noted that strategic customers often received more favourable rates:

“Towards strategic customers, you often have somewhat lower hourly rates than with less strategic customers.”

These examples show how the staffing proposal was shaped by the commercial level at which the opportunity was evaluated. From a project perspective, the strongest estimate might be the one with the highest immediate margin. From an account perspective, however, the strongest estimate could be the one that supported a strategically important customer relationship.

The findings therefore show that estimation involved not only assessing what the project required, but also evaluating how much commercial risk the organisation was willing to accept. This further illustrates how commercial and strategic trade-offs influenced the construction of organisational commitments in practice.

Chapter 6

Discussion

This chapter argues that estimation in time-based IT consulting is best understood as controlled commitment: a practice whose primary product is not a precise prediction, but a defensible commitment under uncertainty. The findings show that estimates must be credible, explainable, and organisationally acceptable in situations where the assignment is not yet fully defined.

This implies an analytical shift from accuracy to defensibility as the operative quality criterion. Estimation is therefore not primarily a question of producing the most accurate forecast, but of establishing a commitment that the organisation can justify to both customers and internal decision-making arenas. This challenges the dominant accuracy-oriented view in software estimation research, where estimates are commonly evaluated in terms of effort, cost, and accuracy [11], and builds on research showing that estimation depends on judgement, uncertainty assessment, and commercial conditions [8, 7].

The four analytical concepts developed in Chapter 3 can therefore be understood as different parts of the same mechanism rather than as separate themes. The estimation basis grounds the commitment by defining what the assignment is assumed to be. Uncertainty handling bounds the commitment by specifying the conditions under which it remains valid. Competence justification legitimises the commitment by making the proposed staffing credible. Commercial and strategic trade-offs align the commitment with organisational and customer-related interests. Together, these practices explain how decision-makers construct commitments that are not necessarily precise, but defensible. This interpretation builds on prior research on estimation, uncertainty, and project success criteria [7, 1].

The discussion is organised around the three research questions. The first section discusses how resource and competence needs are estimated and justified by showing that estimation helps constitute the assignment itself. The second section discusses how uncertainty is handled by showing that time-based contracts relocate commitment rather than remove it. The third section discusses how trade-offs are balanced by showing that estimates are evaluated across multiple organisational levels and that staffing proposals function as both competence claims and risk allocations. The chapter then synthesises these arguments by developing defensibility as the operative quality criterion for estimation in this setting.

6.1 Estimation as constitution of the assignment

The findings point to the way resource and competence needs emerge not from a fixed assignment but through the estimation process itself. In this process, ambiguous and incomplete customer needs are interpreted and transformed into organisational constructs such as roles, competence profiles, seniority levels, workload, hours, and rates.

This implies that estimation cannot be understood as the mere application of judgment to a predefined problem. Instead, estimation emerges as a practice that makes the problem operational and manageable within an organisational context. The estimation basis thus functions not only as input for calculation, but as a foundation that makes the assignment understandable, discussable, and defensible. This aligns with theoretical perspectives that understand estimation as shaped by uncertainty, project assumptions, and commercial considerations [1, 7]. In this study, these dynamics are evident in the way uncertain customer needs are translated into staffing assumptions, competence requirements, and commercial commitments.

Furthermore, the findings challenge the assumption that the object being estimated exists independently of the estimate. Previous research shows that estimation typically relies on expert judgment, historical data, and assessments of project-specific factors such as complexity, team competence, and customer requirements [8, 10]. This study extends this understanding by demonstrating that judgment not only improves or adjusts estimates but also helps define what the

estimate is about. In time-based consulting, professional judgment makes the assignment's content concrete.

An important implication is that the quality of an estimate cannot be assessed as if the assignment existed independently of the estimate. When decision-makers translate ambiguous customer needs into concrete organisational parameters, they simultaneously help define the object that later forms the basis for staffing, pricing, evaluation, and customer expectations. Estimation thus takes on a constitutive function rather than a purely predictive one.

This also has methodological implications for how estimation quality is evaluated. If the assignment and the estimate are developed in parallel, it becomes problematic to assess quality solely by comparing estimated hours with actual billed hours. Such comparisons assume a stable reference point. However, when the assignment is continuously interpreted and shaped during the estimation process, deviations may reflect changes in scope, assumptions, customer decisions, or organisational interpretations rather than poor estimation accuracy.

This does not mean that feedback on estimation accuracy is without value. Prior research indicates that historical data and experience can support improved judgment-based estimation [8]. However, the findings of this study suggest that such feedback has limited explanatory power unless it is interpreted in light of the assumptions, competence choices, and customer conditions that shaped the original estimate. The estimate should therefore be understood not only as a prediction of future work, but as a provisional construction of the assignment itself.

Internal review processes further reinforce this interpretation. In internal evaluations of estimates, the focus is not only on the correctness of the numbers, but also on whether the interpretation of the assignment is credible, commercially viable, and organisationally acceptable. Through these processes, individual judgments are transformed into collective commitments. The estimate becomes defensible not because all uncertainty has been eliminated, but because the organisation has accepted the assumptions and reasoning upon which the commitment is based.

Taken together, the analysis shows that resource and competence needs are estimated and justified through a translation process in which incomplete customer needs are transformed into organisationally actionable categories. Estimation should therefore not be understood merely as preparation for commitment, but as a central practice through which the commitment itself is constructed. Re-

source and competence needs are estimated and justified by producing a defensible version of the assignment, one that can be staffed, priced, reviewed, and communicated before the work is fully known.

6.2 Uncertainty as bounded and redistributed

A key finding of this study is that uncertainty in time-based IT consulting is not eliminated prior to commitment but actively managed through estimation practices. Rather than being reduced or resolved, uncertainty is bounded, communicated, and redistributed in ways that make commitment possible. The findings further show that uncertainty extends beyond technical aspects and should be understood as relational, organisational, and commercial in nature. It may arise from unclear customer needs, shifting priorities, an unstable scope, incomplete information, customer dependencies, or uncertainty about future resource availability.

Prior research shows that uncertainty concerns not only identifiable risks, but also assumptions, objectives, priorities, design, logistics, and relationships among project actors [23]. The findings further show that uncertainty is managed through practices such as assumptions, ranges, reservations, qualifications, and phased delivery. These practices make uncertainty visible while also defining the conditions under which the estimate is valid. In this sense, handling uncertainty is not primarily about achieving greater precision, but about making uncertainty manageable enough to enable commitment. Decision-makers do not necessarily remove uncertainty; rather, they transform it into a form that can be discussed, qualified, and used as a basis for action.

In this context, assumptions and reservations play a load-bearing role in the estimate. They are not peripheral disclaimers but central mechanisms for defining what the supplier is willing to commit to, and under which conditions. A reservation does not merely express a lack of knowledge; it also establishes a defensible position if the project evolves differently than expected. In this way, uncertainty is incorporated into the estimate as a set of explicit conditions for commitment.

Phased delivery further reinforces this logic. Discovery phases, proofs of concept, and limited initial engagements enable the supplier to enter into a customer relationship without committing to a full scope before the assignment is suffi-

ciently understood. Such phases reduce uncertainty by generating additional information, while simultaneously distributing commitment over time. Prior research supports the broader view that uncertainty management involves clarification, assumptions, and flexibility in the face of incomplete information [23, 1]. The findings of this study extend this perspective by showing that phased delivery also sequences commitment: the supplier can commit to learning before committing to full delivery.

Furthermore, the findings show that time-based contracts do not eliminate supplier commitment. Formally, such contracts may limit responsibility by requiring the customer to pay for time and competence rather than for a predefined outcome. However, the findings indicate that estimates still create expectations. Customers may use estimates as a basis for planning, budgeting, and subsequent evaluation, meaning that the supplier remains implicitly committed through the estimate, the customer relationship, and reputational considerations.

The key analytical point is therefore not that time-based contracts involve less commitment than fixed-price contracts, but that commitment is embedded differently. In fixed-price projects, commitment is more visibly tied to contractual scope, price, and delivery obligations. In time-based work, commitment is more often located in the estimate, the associated assumptions, and the relationship that forms around it. Assumptions, reservations, ranges, and phased delivery thus perform boundary-setting work that the contract itself does not fully provide.

This implies that time-based contracts do not simply reduce commitment, but rather relocate it. Prior research has shown that estimation and bidding are shaped by uncertainty, margins, client budgets, competitive pricing, and long-term strategic considerations [7, 9]. The findings of this study extend this perspective by showing that such considerations are not only influential in shaping the estimate but also integral to how organisations determine what kind of commitment they can defend.

Taken together, the analysis shows that uncertainty is handled through practices that make commitment conditional, bounded, and revisable. Uncertainty is not eliminated prior to commitment; rather, it is incorporated into the estimate through assumptions, reservations, ranges, and phases that define what the organisation can reasonably stand behind. The answer to the second research question is, therefore, that uncertainty is managed by converting it into explicit

conditions for commitment and by redistributing it across assumptions, phases, customer expectations, and organisational responsibility.

6.3 Estimation as negotiation between organisational considerations

The findings of this study show that trade-offs between competence, availability, cost, and commercial considerations are not resolved through optimisation alone. Instead, they are balanced through organisational judgement when constructing a defensible offer. Staffing and pricing decisions are therefore not direct outputs of a technical estimation basis but the result of reconciling multiple, sometimes competing, organisational considerations.

Decision-makers must account for delivery quality, competence requirements, seniority, resource availability, cost, margins, customer expectations, and strategic value. While these considerations may occasionally align, they often conflict. For example, a senior-heavy team may enhance delivery credibility and reduce uncertainty, yet increase costs and weaken commercial competitiveness. Conversely, a more available or lower-cost team may be easier to staff and sell, but may require stronger assumptions, more support, or a more limited commitment.

These findings support prior research on project-based organisations and resource allocation, which shows that staffing decisions are shaped by availability constraints, competing priorities, temporary project structures, and dynamic organisational conditions [6, 14]. However, this study extends previous insights by demonstrating that such constraints are not confined to the delivery phase. They are already present during estimation, where the proposed team plays a central role in making the offer both credible and actionable.

Moreover, the findings indicate that the same estimate may be evaluated differently depending on the organisational level at which it is considered. At the project level, estimates are typically assessed in terms of effort, cost, margin, delivery risk, and the adequacy of the proposed team. At the account or customer-relationship level, evaluation may instead focus on future opportunities, strategic positioning, and long-term customer value. As a result, a proposal that appears weak from a narrow project-margin perspective may still be defensible when viewed in light of its strategic importance.

This implies that an estimate can simultaneously be perceived as weak and defensible, depending on the evaluative frame applied. A proposal with low immediate margins may be unattractive at the project level, yet justified at the account level if it strengthens access to a strategically important customer. Consequently, there is no single criterion for what constitutes a “good” estimate in this context. Rather, the quality of an estimate depends on the organisational lens through which it is assessed.

Accordingly, an estimate must be defensible across multiple evaluative frames. It must be technically feasible, commercially viable, realistically staffable, and aligned with the broader customer relationship. This supports the theoretical framing of estimation as a negotiated compromise rather than a purely technical outcome, building on research on commercial estimation and competence in project settings [7, 13].

The findings further demonstrate that estimation and resource allocation are closely intertwined. Although analytically distinct, these processes are deeply interconnected in practice. When a team is proposed within an estimate, the organisation is effectively anticipating a future allocation of resources. Roles and seniority levels create expectations about resource commitments that have not yet been formally assigned. Estimation thus functions as a forward-looking organisational practice in which commitments begin to take shape prior to formal allocation decisions.

Resource availability therefore operates as a prospective constraint on estimation, not merely as a retrospective constraint on allocation. By including a proposed team in an estimate, the organisation makes an implicit preliminary commitment regarding resource use. Estimation and allocation are thus not strictly sequential, but partially overlapping processes. The estimate anticipates an allocation that the organisation must subsequently attempt to realise. Existing research on resource allocation shows that staffing decisions are shaped by availability constraints, competing priorities, temporary project structures, and dynamic organisational conditions [6, 14]. This study extends that perspective by shifting the analytical focus upstream to the pre-sales phase, where future staffing constraints are already translated into customer-facing commitments.

Seniority and team composition play a particularly important role in this process. The findings show that seniority is not solely a matter of competence and cost, but also functions as a mechanism for distributing uncertainty and delivery

risk. Senior consultants may be included to mitigate ambiguity, manage customer interactions, and enhance robustness under uncertainty. More junior profiles may be appropriate when tasks are clearer, scope is limited, or senior support is available.

Team composition can thus be interpreted as a mechanism for allocating delivery risk. A senior-heavy team may signal that the supplier anticipates ambiguity, coordination challenges, or intensive customer interaction. A more junior-heavy team may be defensible when work is standardised, clearly defined, or supported by senior oversight. Prior research highlights the importance of competence, seniority, and knowledge-intensive work in project and professional service settings [13, 22]. The findings of this study add that the mix of seniority also reflects how uncertainty is distributed within the delivery model and how much risk the supplier is willing to absorb.

Consequently, competence justification serves both a delivery function and a risk function. Roles, competence profiles, and seniority levels make the offer visible and intelligible to the customer while simultaneously signalling how uncertainty is managed. This is particularly important in knowledge-intensive services, where the quality and value of work can be difficult for customers to assess in advance [22]. Competence must therefore be made explicit and defensible by structuring and justifying roles.

Taken together, the findings show that trade-offs between competence, availability, cost, and commercial considerations are not resolved through optimisation, but through organisational judgement. Estimation involves constructing an offer in which competing considerations are rendered mutually defensible within a single customer-facing proposal. The answer to the third research question is therefore that trade-offs are balanced by developing an offer that can be justified across multiple organisational logics simultaneously: delivery feasibility, resource availability, commercial viability, and strategic value.

6.4 Defensibility as the operative quality criterion

An overarching insight from this study is that estimation in time-based IT consulting is governed less by accuracy than by defensibility, the operative quality criterion. Estimates must serve as decision-making bases in contexts characterised

by uncertainty and incomplete information, while remaining usable across multiple arenas, including customer dialogue, internal review, staffing discussions, pricing, and commercial evaluation.

This perspective challenges the dominant accuracy-oriented view within software estimation research. Existing studies have produced significant knowledge regarding estimation methods, expert judgement, the use of historical data, and approaches to accuracy evaluation [11, 8]. However, the analysis presented here indicates that accuracy alone is insufficient as a quality criterion in situations where the assignment is still under construction, where multiple actors rely on the estimate, and where commitment must be justified before the full scope of work is known.

Within such conditions, greater precision does not necessarily imply higher quality. A single-point estimate may appear clear while obscuring uncertainty and creating stronger expectations than the organisation can reasonably defend. By contrast, estimates expressed through ranges, assumptions, reservations, or phased structures may appear less precise, yet provide a more defensible basis for action. Defensibility therefore captures the estimate's capacity to be credible, explainable, and organisationally acceptable under conditions of uncertainty.

Rather than representing an additional dimension alongside accuracy, defensibility functions as an integrating criterion. An estimate can be considered defensible when the assignment has been rendered sufficiently understandable, when uncertainty has been bounded and made explicit, when competence choices can be justified, and when commercial trade-offs are articulated in a coherent manner. In this way, defensibility links the study's analytical concepts into a unified mechanism, illustrating how organisations move from incomplete customer needs to commitments they are prepared to uphold.

It is important to emphasise that this does not imply that accuracy is irrelevant. Prior research provides important insights into estimation accuracy, expert judgement, and uncertainty in project work [11, 8, 23]. However, the findings of this study suggest that accuracy is insufficient as the primary evaluative criterion in this setting. A highly precise estimate may still be problematic if it conceals uncertainty or generates unjustified expectations. Conversely, a less precise estimate may be more robust if it explicitly incorporates uncertainty through ranges, assumptions, reservations, or phased commitments. The critical issue is therefore not solely whether the estimate ultimately proves correct, but whether it was

reasonable, transparent, and responsibly bounded at the time the commitment was made.

This perspective also clarifies the relationship between estimation, contract type, and risk. In time-based arrangements, formal contractual responsibility may be limited; however, the estimate continues to shape expectations and implies a form of responsibility. Commitment is therefore not eliminated but relocated—from the formal contract into the estimate itself and the relational context surrounding it. The estimate becomes a central arena in which responsibility is established, negotiated, and legitimised prior to full knowledge of the work involved.

Seen in this light, this study conceptualises estimation in time-based IT consulting not as a process aimed primarily at producing precise predictions, but as a practice of constructing defensible commitments under uncertainty. The concept of controlled commitment captures how decision-makers translate incomplete customer needs into staffing and competence configurations, how they bound uncertainty through assumptions and phased approaches, and how they balance delivery feasibility, resource availability, cost, margin, and strategic considerations within a single, customer-facing offer.

This implies a fundamental shift in how estimation should be conceptualised: away from the pursuit of maximal accuracy and towards an understanding of what the organisation can responsibly defend.

6.5 Implications for practice

The findings have several implications for how organisations can understand and improve estimation practices in time-based IT consulting.

First, organisations should treat estimation as a strategic and organisational practice, not as an isolated calculation activity. This means strengthening the processes through which customer needs are translated into internal categories such as roles, competence profiles, assumptions, and staffing expectations. Involving relevant experts early, actively applying experience-based knowledge, and discussing alternative interpretations of the assignment can help create more robust estimates. At the same time, organisations may benefit from linking earlier estimates to later delivery outcomes, not only to improve precision but also to support

organisational learning about assumptions, uncertainty, and commitment.

Second, organisations should treat assumptions and reservations as load-bearing parts of the offer. Since commitment in time-based work is partly based on estimates rather than the contract, the assumptions underlying those estimates are crucial. They define what the supplier is actually committing to. Assumptions should therefore be specific enough to be evaluated later, and reservations should be formulated so they can be invoked if relevant conditions change.

Third, organisations should use phased delivery as a commitment staging device, not only as a technical or methodological choice. Discovery phases, proofs of concept, and limited initial engagements can reduce uncertainty by creating new information, but they also limit the organisation's early exposure. They allow the supplier to build knowledge before taking on stronger commitments. The choice between a phased and a full-scope offer should therefore be understood as a question of how much commitment the organisation is willing to make, based on the current estimate.

Fourth, organisations should connect estimation and staffing more explicitly. Estimation already contains an implicit allocation of future resources, even before named individuals are formally assigned. Organisations may therefore benefit from closer coordination between estimation, sales, delivery, and staffing functions. This includes making visible how availability, competence, seniority, and strategic customer considerations influence the estimate.

Fifth, organisations should make explicit which level of commercial evaluation is being used in internal review. If a proposal is justified by project-level economics, it should be assessed on project-level assumptions about effort, cost, margin, and delivery risk. If it is justified by an account-level strategy, it should be made explicit and evaluated accordingly. Otherwise, strategic value may become an unexamined justification for weak project-level estimates.

Finally, organisations should recognise that team composition is also a way of handling risk. Seniority and competence are not only questions of skill and cost; they also shape how uncertainty is absorbed by the supplier and the customer. Being explicit about this can improve both internal review and customer communication.

Overall, the practical implication is that organisations should develop estimation practices that prioritise credibility, transparency, and responsible handling of uncertainty over apparent precision.

Chapter 7

Conclusion

This study has examined how resource and competence needs are estimated in time-based IT consulting work, with particular attention to how uncertainty is handled and how staffing decisions are shaped. Based on the analysis, the findings show that estimation cannot primarily be understood as a technical or predictive activity. Rather, estimation is an organisational practice through which decision-makers construct, bound, and negotiate commitments under uncertainty.

The main contribution of the study is the concept of controlled commitment as an analytical framework for understanding estimation in this context. Controlled commitment captures how organisations commit enough to make a customer opportunity actionable, while limiting exposure to unclear scope, unstable customer needs, uncertain competence requirements, future resource availability, and commercial risk. This means that estimation is not primarily about producing accurate forecasts, but about establishing commitments that appear credible, manageable, and defensible before the assignment is fully defined.

First, the analysis shows that estimation involves constructing the assignment. Customer needs are translated from incomplete or ambiguous descriptions into organisational categories that make them actionable, such as roles, competence profiles, seniority levels, workload, hours, rates, and assumptions. The assignment is therefore not simply an input to estimation, but is partly produced through the estimation process itself. Estimation is thus constitutive rather than merely predictive: it helps define what the organisation later staffs, prices, reviews, and communicates to the customer.

Second, the study shows that uncertainty is rarely eliminated. Instead, it is

bounded, communicated, and redistributed through assumptions, ranges, reservations, and phased approaches. These practices make uncertainty visible and manageable, while also defining the conditions under which the organisation is willing to commit. In this way, uncertainty is not removed before action is taken but structured to make action possible. A central finding is that time-based contracts do not remove supplier commitment. Rather, they shift commitment from the formal contract to the estimate, its attached assumptions, and the customer relationship surrounding it.

Third, the findings show that staffing and pricing cannot be understood in isolation. They are shaped through trade-offs between delivery quality, competence, capacity, cost, profitability, and strategic considerations. The same estimate may therefore be evaluated differently depending on whether it is assessed from a project-level or customer-relationship perspective. A proposal may be weak at the project level, but still defensible at the account level if it supports a strategically important customer relationship. Estimation is thus not only a calculation of expected work, but a negotiation between competing organisational considerations.

The findings also show that estimation and allocation are not fully sequential processes. When a team is proposed in an estimate, the organisation is already anticipating a future allocation that must be realised later. Availability therefore functions not only as a constraint on allocation, but also as a prospective constraint on estimation. Similarly, seniority and team composition are not only questions of competence and cost. They also serve as mechanisms for distributing delivery risk by signalling how much uncertainty the supplier is willing to absorb and how that uncertainty will be handled under the proposed delivery model.

Taken together, the study shows that the four analytical concepts used in the thesis are best understood as parts of a single mechanism. The estimation basis grounds the commitment by defining what the assignment is assumed to be. Uncertainty handling bounds the commitment by specifying the conditions under which it remains valid. Competence justification legitimises the commitment by making the proposed staffing credible. Commercial and strategic trade-offs align the commitment with organisational and customer-related interests. Together, these practices explain how decision-makers translate incomplete customer needs into a customer-facing offer that the organisation can defend.

On this basis, the study challenges established understandings of estimation

in software development, where accuracy often appears as the central quality criterion. In time-based and knowledge-intensive contexts, defensibility is a more relevant criterion. Estimates are used in decision situations characterised by uncertainty, multiple stakeholders, and assignments that are not yet fully clarified. Their quality therefore depends not only on whether they later prove accurate but also on whether they can be justified at the time they are made.

Overall, the study shows that estimation functions as a link between uncertain demand and organisational action. It enables organisations to act under uncertainty, not by eliminating uncertainty, but by structuring it and making it manageable. The concept of controlled commitment therefore contributes to understanding estimation as a relational and organisational practice, particularly in contexts where knowledge, uncertainty, and responsibility are closely intertwined. This opens the door to further research into how commitment is constructed in other consulting contexts and how customers interpret estimates, assumptions, reservations, and phased commitments as bases for expectations and accountability.

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Appendix A

Sikt notification form (Meldeskjema)



Meldeskjema for behandling av personopplysninger

Referansenummer

290155

Hvilke personopplysninger skal du behandle?

- Stemme på lydopptak

Behandlingsansvar

Behandlingsansvarlig institusjon

Universitetet i Stavanger / Det teknisk- naturvitenskapelige fakultet / Institutt for sikkerhet, økonomi og planlegging

Prosjektansvarlig

Sindre Lorentzen, sindre.lorentzen@uis.no, tlf: 0

Er behandlingsansvaret delt med flere institusjoner?

Nei

Prosjektinformasjon

Tittel

Master - Industrial economics

Sammendrag

Formålet med dette masterprosjektet er å undersøke hvordan usikkerhet håndteres i estimering av IT-prosjekter i konsulentorganisasjoner. Studien fokuserer spesielt på hvordan beslutningstakere vurderer kompetanse, ressursbehov og prosjektomfang i tidlige faser av prosjektplanlegging, hvor informasjon ofte er begrenset og usikkerheten høy. Prosjektet gjennomføres som en kvalitativ casestudie av estimeringspraksis i konsultentselskapet Sopra Steria. Datagrunnlaget består av semistrukturerte intervjuer med ansatte som arbeider med prosjektledelse, estimering og bemanning av IT-prosjekter. Formålet med intervjuene er å få innsikt i hvordan estimerer utvikles i praksis, hvilke faktorer som påvirker beslutningene og hvilke organisatoriske mekanismer som brukes for å håndtere usikkerhet. Studien vil bidra til økt forståelse av hvordan estimering og kompetanseallokering foregår i kunnskapsintensive organisasjoner. Resultatene kan gi innsikt i hvordan konsulentorganisasjoner håndterer usikker

Akademisk nivå

Master

Kontaktinformasjon, student

Torjus Jåstad Knudsen, tj.knudsen@stud.uis.no, tlf: 93865329

Overordnet formål med behandlingen

Forskning eller studentoppgave

Beskriv hva du skal forske på, og hvorfor du må behandle personopplysninger for å oppnå det angitte formålet

Dette masterprosjektet undersøker hvordan usikkerhet håndteres i estimering av IT-prosjekter i en konsulentorganisasjon. Studien fokuserer på hvordan beslutningstakere vurderer kompetanse, ressursbehov og prosjektomfang i tidlige faser av prosjektplanlegging. Datainnsamlingen gjennomføres gjennom semistrukturerte intervjuer med ansatte som arbeider med prosjektledelse, estimering og bemanning av IT-prosjekter. I forbindelse med intervjuene vil det behandles begrensede personopplysninger, som navn og stilling, for å kunne rekruttere relevante informanter og dokumentere deres rolle i studien. Opplysningene vil kun brukes til forskningsformål og anonymiseres i den endelige oppgaven.

Totalt antall registrerte i prosjektet

1-99

Lovlig grunnlag for å behandle alminnelige kategorier personopplysninger (GDPR art. 6)

- Samtykke

Supplerende lovlig grunnlag**Tillatelser**

Utvalg 1

Beskriv utvalget

Utvalget består av ansatte i konsultentselskapet Sopra Steria som arbeider med prosjektledelse, estimering eller bemanning av IT-prosjekter. Disse personene har erfaring med planlegging og estimering av prosjekter i konsulentorganisasjoner. Utvalget vil bestå av cirka 5–8 personer.

Beskriv hvordan du finner frem til eller kontakter utvalget

Deltakerne rekrutteres gjennom profesjonelt nettverk i Sopra Steria. Potensielle informanter kontaktes direkte via e-post eller muntlig forespørsel med informasjon om prosjektet og forespørsel om deltakelse.

Aldersgruppe

30 - 60

Hvilke personopplysninger vil bli behandlet om utvalg 1?

- Stemme på lydopptak

Lovlig grunnlag for å behandle alminnelige kategorier personopplysninger (GDPR art. 6)

- Samtykke

Hvordan innhentes opplysningene om utvalg 1?

Personlig intervju

Oppsummer tematisk innhold i intervjuene, og hvordan de praktisk gjennomføres

Intervjuene vil være semistrukturerte og fokusere på erfaringer med estimering av IT-prosjekter, håndtering av usikkerhet, vurdering av kompetansebehov og beslutninger knyttet til bemanning av prosjekter. Intervjuene gjennomføres digitalt eller fysisk, varer omtrent 30–60 minutter, og tas opp med lyd etter samtykke fra deltakerne. Opptak av intervjuene vil gjøres ved bruk av Nettskjema sin løsning for sikker lydopptak.

Vedlegg

[Tusen takk for at du ønsker å delta i intervjuet.docx](#)

Informasjon til utvalg 1

Mottar utvalget informasjon om behandlingen av personopplysningene?

Ja

Hvordan mottar utvalget informasjon om behandlingen?

Skriftlig (papir eller elektronisk)

Informasjonsskriv

[informasjonsskriv.docx](#)

Tredjepersoner

Innhenter prosjektet informasjon om tredjepersoner?

Nei

Dokumentasjon

Hvordan dokumenteres samtykkene?

- Manuelt (papir)

Hvordan kan samtykket trekkes tilbake?

Samtykke kan trekkes tilbake ved å kontakte prosjektleder på e-post.

Hvordan kan de registrerte få innsyn, rettet eller slettet personopplysninger om seg selv?

Deltakerne kan be om innsyn i, retting eller sletting av personopplysninger ved å kontakte prosjektleder på e-post.

Sikkerhetstiltak

Vil direkte personidentifiserende opplysninger lagres atskilt fra øvrige data ?

Ja

Hvilke tekniske og fysiske tiltak sikrer personopplysningene?

- Adgangsbegrensning

Hvor blir personopplysningene behandlet?

- Maskinvare

Mottakere

Hvem har tilgang til personopplysningene?

- Student (studentprosjekt)

Overføres personopplysninger til et tredjeland?

Nei

Avslutning

Prosjektperiode

16.03.2026 - 01.06.2026

Hva skjer med dataene ved prosjektslutt?

Data slettes (sletter rådataene)

Vil enkeltpersoner kunne gjenkjennes i publisasjon?

Nei

Tilleggsopplysninger

Personvernrisiko

Vil behandlingen innebære høy personvernrisiko slik at det er behov for en DPIA?

Vet ikke

Appendix B

Information sheet and consent form

Informasjonsskriv til deltaker i forskningsprosjekt

Vil du delta i forskningsprosjektet “Decision-Making in the Estimation of Resource and Competency Needs in IT Consulting Projects: A Qualitative Case Study at Sopra Steria”?

Formål

Jeg heter Torjus Jåstad Knudsen og er masterstudent i industriell økonomi ved Universitetet i Stavanger. Dette intervjuet gjennomføres som en del av min masteroppgave.

Formålet med prosjektet er å undersøke hvordan estimering av ressurs- og kompetansebehov gjennomføres i IT-konsulentprosjekter, og hvilke organisatoriske mekanismer som påvirker disse beslutningene. Studien ser særlig på hvordan usikkerhet håndteres i tidlige faser av prosjektplanlegging.

Hvorfor spør vi deg om å delta?

Du får denne forespørselen fordi du arbeider med eller har erfaring med estimering, bemanning eller prosjektledelse i IT-konsulentprosjekter. Deltakere rekrutteres gjennom profesjonelt nettverk i Sopra Steria.

Hvem er ansvarlig?

Universitetet i Stavanger er ansvarlig for behandlingen av personopplysninger i prosjektet.

Hva innebærer det å delta?

Deltakelse innebærer et semistrukturert intervju på omtrent 45–60 minutter, hvor du blir spurt om dine erfaringer og refleksjoner knyttet til estimering, prosjektplanlegging og bemanning i IT-prosjekter.

Intervjuet kan gjennomføres digitalt eller fysisk. Dersom du samtykker, vil intervjuet bli tatt opp som lydopptak ved bruk av Nettskjema sin løsning for sikkert lydopptak.

Opplysninger som kan behandles i prosjektet inkluderer:

- navn
- stilling eller rolle
- erfaring knyttet til prosjektarbeid

I den ferdige oppgaven vil alle personopplysninger anonymiseres slik at enkeltpersoner ikke kan identifiseres

Frivillig å delta

Det er frivillig å delta i prosjektet. Det vil ikke ha noen negative konsekvenser for deg hvis du ikke vil delta, eller hvis du senere velger å be om å få dine personopplysninger slettet.

Dine rettigheter

Så lenge du kan identifiseres, har du rett til å be om innsyn, retting eller sletting av opplysninger om deg. Du kan også klage til Datatilsynet. Les mer her: <https://www.datatilsynet.no/rettigheter-og-plikter/den-registrertes-rettigheter/>.

Bruk av personopplysninger

Vi vil bare bruke opplysningene om deg til formål vi har fortalt om i dette skrivet. Vi behandler personopplysningene konfidensielt og i samsvar med personvernregelverket.

Opplysningene vil kun bli brukt til forskningsformål i forbindelse med denne masteroppgaven. Data behandles konfidensielt og lagres sikkert. Kun studenten og veileder ved Universitetet i Stavanger vil ha tilgang til datamaterialet.

Lovlig grunnlag for behandling av personopplysninger

Vi behandler opplysningene om deg basert på ditt samtykke. Du står fritt til å trekke ditt samtykke tilbake.

Hva skjer med personopplysningene dine etter prosjektet er avsluttet?

Prosjektet planlegges avsluttet **01.06.2026**. Etter at prosjektet er ferdigstilt vil lydopptak og personopplysninger bli slettet.

Kontaktinformasjon

- Prosjektansvarlig: Torjus Jåstad Knudsen, Masterstudent, Universitetet i Stavanger, tj.knudsen@uis.no
- For spørsmål om personvern i forskningsprosjektet kan du kontakte personvernombudet ved Universitetet i Stavanger.
- Sikt – Kunnskapssektorens tjenesteleverandør har vurdert at behandlingen av personopplysninger i dette prosjektet er i samsvar med personvernregelverket. For spørsmål om vurderingen: personverntjenester@sikt.no

Samtykke til å delta

Jeg har mottatt informasjon om prosjektet [tittel] og samtykker til å delta.

Signatur: _____ Dato: _____

Appendix C

Interview guide

Tusen takk for at du ønsker å delta i dette intervjuet.

Jeg er masterstudent i Industriell økonomi ved Universitetet i Stavanger, og intervjuet gjennomføres som en del av min masteroppgave. Studien undersøker hvordan beslutningstakere opplever og begrunner estimering av ressurs- og kompetansebehov i timebaserte IT-konsulentprosjekter.

Intervjuet varer omtrent 45–60 minutter. Samtalen tas opp dersom du samtykker, for å sikre korrekt gjengivelse. Alle opplysninger anonymiseres. Deltakelse er frivillig, og du kan når som helst trekke deg eller velge å ikke svare på spørsmål. Data brukes kun til denne masteroppgaven og slettes etter at prosjektet er avsluttet.

Har du spørsmål før vi starter?

Faktaspørsmål

For å forstå konteksten bedre ønsker jeg først å stille noen korte bakgrunns spørsmål:

- Hva er din rolle i organisasjonen?
- Hvor lenge har du hatt beslutningsansvar knyttet til bemanning eller estimering?
- Hvilke typer prosjekter er du vanligvis involvert i?
- Hvor ofte deltar du i estimeringsprosesser?

Introduksjonsspørsmål

Formålet her er å få en åpen beskrivelse av praksis.

- Kan du beskrive hvordan estimering av ressurs- og kompetansebehov typisk foregår i et nytt prosjekt?
- Bruker dere tidligere prosjekter eller erfaringstall som referanse når dere estimerer?
- Bruker dere faste modeller, maler eller mer erfaringsbaserte vurderinger?
- Vurderer dere flere mulige utfall, for eksempel optimistisk, mest sannsynlig og pessimistisk?
- Kan du fortelle om en konkret situasjon der du var involvert i en slik vurdering?
- Hvordan opplevde du beslutningsprosessen i den situasjonen?

Overgangsspørsmål

Her går vi fra beskrivelse til refleksjon.

- Hvordan tenker du når du skal vurdere hvilke ressurser som trengs?
- Hvilke faktorer vurderer du først?
- Hvordan begrunner du dine vurderinger overfor andre?
- Hvem involveres i beslutningen, og hvordan påvirker det prosessen?
- Hva vil du si kjennetegner en god estimering?

Nøkkelspørsmål

1. Organisatoriske mekanismer

- Hvordan opplever du at organisasjonsstruktur og rollefordeling påvirker estimeringsbeslutninger?
- Hvem har mest innflytelse i slike beslutninger?
- Hvordan påvirker samspillet mellom ulike funksjoner (for eksempel salg og leveranse) vurderingene?
- Hva mener du forklarer hvorfor estimeringsprosessen foregår slik den gjør?

2. Usikkerhet

- Hvordan opplever du usikkerhet i estimeringsfasen?
- Hva skaper mest usikkerhet?
- Hvordan håndterer du situasjoner der informasjonen er ufullstendig?
- Opplever du at tidlige antakelser eller de første tallene som nevnes ofte påvirker resten av vurderingen?
- Hender det at estimerer blir mer optimistiske enn det faglige grunnlaget egentlig tilsier?
- Hvordan prøver du eventuelt å unngå slike skjevheter i vurderingene?
- Hvordan begrunner du beslutninger når det er høy grad av usikkerhet?

3. Prioriteringer og avveininger

- Hvilke hensyn må du typisk avveie når du estimerer bemanning?

- Hvordan balanserer du optimal kompetanse mot tilgjengelig kapasitet?
- Hvordan påvirker økonomiske hensyn dine vurderinger?
- Kan du beskrive en situasjon der ulike hensyn sto i konflikt, og hvordan du begrunnet valget?

Kompliserte og sensitive spørsmål

- Opplever du press i estimeringsprosessen, enten internt eller fra kunde?
- Finnes det situasjoner hvor estimering påvirkes av kommersielle eller strategiske hensyn?
- Hender det at estimerer tilpasses det som er kommersielt mulig å selge, heller enn det som ideelt sett virker mest realistisk?
- Hvordan håndterer dere spenningen mellom realistiske estimerer og det å vinne oppdrag?
- I hvilken grad påvirker økonomiske mål hvordan beslutninger tas?
- Har du opplevd at beslutninger tas på andre grunnlag enn det som fremstår som faglig optimalt?

Avslutning

- Er det noe vi ikke har snakket om som du mener er viktig for å forstå estimeringsprosessen?
- Er det noe du ønsker å utdype eller legge til?
- Er det andre du mener jeg bør intervju?

Tusen takk for at du tok deg tid til å delta.

Appendix D

Theme development

Table D.1: Example of theme development from interview excerpt to final theme

Data excerpt	Initial code	Descriptive category	Final theme
<i>P2</i> : The customer request may be very thin or very detailed. Then it is about interpreting what needs to be solved and what type of competence is required.	Interpreting what must be solved; identifying required competence	Customer need, scope, context, and incomplete information	Translating incomplete customer needs into organisationally defensible staffing commitments
<i>P4</i> : We start with what we clearly know, what we have done before, and what we feel confident about. Then we discuss what we do not know and what may become a factor.	Identifying knowns and unknowns in the estimate	Risk, uncertainty, and risk management	Bounding uncertainty through assumptions, ranges, and staged commitment
<i>P6</i> : You always try not to staff the most available resource, but the one who is best suited to carry out the project.	Choosing fit rather than only availability	Resource, competence, and candidate assessment	Negotiating staffing between fit, availability, margin, and strategic value

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